Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If three positive real numbers a, b, c are in A.P. such that $abc = 4$, then the minimum possible value of b is

(A) $2^{3/2}$

(B) $2^{2/3}$

(C) $2^{1/3}$

(D) $2^{5/2}$

Q.2

If $(1 + 3 + 5 + \ldots + p) + (1 + 3 + 5 + \ldots + q)$
$= (1 + 3 + 5 + \ldots + r)$
where each set of parentheses contains the sum of consecutive odd integers as shown, what is the smallest possible value of $(p + q + r)$ where $p > 6$?

(A) 12

(B) 21

(C) 45

(D) 54

Q.3

$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \ldots + \frac{1}{n(n+1)}$ equals

(A) $\frac{1}{n(n+1)}$

(B) $\frac{n}{n+1}$

(C) $\frac{2n}{n+1}$

(D) $\frac{2}{n(n+1)}$

Q.4

It is given that $\frac{1}{2^n \sin\alpha}, 1, 2^n \sin a$ are in A.P. for some value of α. Let say for $\mathrm{n} = 1$, the α satisfying the above A.P. is α_1, for $\mathrm{n} = 2$, the value is α_2 and
so on. If $\mathrm{S} = \sum_{i=1}^{\infty} \sin\alpha_i$, then the value of S is

(A) 1

@IITJEE_Advanced

(B) $\frac{1}{2}$

(C) 2

(D) None of these

Q.5

If $x = 1 + a + a^2 + \ldots\ldots\ldots\ldots\ldots\ldots$ infinity and
$y = 1 + b + b^2 + \ldots\ldots\ldots\ldots\ldots\ldots$..to infinity, where a, b are proper fractions, then $1 + ab + a^2b^2 + \ldots$ to infinity is equal :

(A) $\frac{xy}{x+y-1}$

(B) $\frac{xy}{x-y-1}$

(C) $\frac{xy}{x-y+1}$

(D) $\frac{xy}{x+y+1}$

Q.6

The sum of the series $1 + \frac{1.3}{6} + \frac{1.3.5}{6.8} + \ldots\ldots\ldots\ldots \infty$ is

(A) 1

(B) 0

(C) ∞

(D) 4

Q.7

For the series $\sum t_n, S_n = 3t_n - 2$
where $S_n = \sum_{n=1}^{n} t_n$. Then $\sum_{n=1}^{n} S_n$

(A) $6\left(\frac{3^n}{2^n} - 1\right) - 2n$

(B) $6^n - 1 - 2n$

(C) $2n - 6\left(\frac{2^n}{3^n} + 1\right)$

(D) None of these

Q.8

The 20th terms of the series $2 + 3 + 5 + 9 + 16$
$+ \ldots\ldots$ is

(A) 950

(B) 975

@IITJEE_Advanced

(C) 990

(D) 1010

Q.9

An A.P. whose first term is unity and in which the sum of first half of any even number of terms to that of second half of the same number of terms is a constant ratio, then the common difference is :

(A) 2

(B) 1

(C) 3

(D) None of these

Q.10

The sum of $\frac{\frac{1}{2}\cdot\frac{2}{2}}{1^3}+\frac{\frac{2}{2}\cdot\frac{3}{2}}{1^3+2^3}+\frac{\frac{3}{2}\cdot\frac{4}{2}}{1^3+2^3+3^3}+\ldots\ldots\ldots$ upto n terms is equal to

(A) $\frac{n-1}{n}$

(B) $\frac{n}{n+1}$

(C) $\frac{n+1}{n+2}$

(D) $\frac{n+1}{n}$

Q.11

Let a, b, c, be in A.P. with a common difference d. Then $e^{1/c}, e^{b/ac}, e^{1/a}$ are in :

(A) G.P. with common ratio e^{d}

(B) G.P with common ratio $e^{1/d}$

(C) G.P. with common ratio $e^{d/\left(b^2-d^2\right)}$

(D) A.P.

Q.12

The least value of n (a natural number), for which the sum S of the series $1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\ldots\ldots$ differs from S_n by a quantity $<10^{-6}$, is

(A) 21

(B) 20

(C) 19

@IITJEE_Advanced

(D) None

Q.13

$\sum_{k=1}^{n} k(1+1/n)^{k-1} =$

(A) $n(n-1)$

(B) $n(n+1)$

(C) n^2

(D) $(n+1)^2$

Q.14

What does the series
$1 + 3^{-\frac{1}{2}} + 3 + \frac{1}{3\sqrt{3}} + \ldots$ represents?

(A) AP

(B) GP

(C) HP

(D) None of the above series

Q.15

In a, G.P. of $3n$ terms, S_1 denotes the sum of first n terms, S_2 the sum of the second block of n terms and S_3^2 the sum of last n terms. Then S_1, S_2, S_3, are in

(A) A.P.

(B) GP.

(C) H.P.

(D) None of these

Q.16

$fx > 0, \frac{x^n}{1+x+x^2+\ldots+x^{2n}}$ is

(A) $\leq \frac{1}{2n+1}$

(B) $< \frac{2}{2n+1}$

(C) $\geq \frac{1}{2n+1}$

(D) $> \frac{2}{2n+1}$

Q.17

@IITJEE_Advanced

If the A.M. between a and b is m times their H.M. then $a : b =$

(A) $\sqrt{m}+\sqrt{m-1} : \sqrt{m}-\sqrt{m-1}$

(B) $\sqrt{m}-\sqrt{m-1} : \sqrt{m+1}+\sqrt{m-1}$

(C) $\sqrt{m}+\sqrt{m+1} : \sqrt{m}-\sqrt{m+1}$

(D) None of these

Q.18

If a, b, c are positive numbers, then least value of $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)$ is

(A) 1

(B) 6

(C) 9

(D) None

Q.19

If $a+b+c=3$ and $a>0, b>0, c>0$, then the greatest value of $\mathrm{a}^2\ \mathrm{b}^3\mathrm{c}^2$ is

(A) $\frac{3^{10}\cdot 2^4}{7^7}$

(B) $\frac{3^9\cdot 2^4}{7^7}$

(C) $\frac{3^8\cdot 2^4}{7^7}$

(D) None of these

Q.20

If a, b, c are the sides of a triangle, then the minimum value of $\frac{a}{b+c-a}+\frac{b}{c+a-b}+\frac{c}{a+b-c}$ is equal to

(A) 3

(B) 6

(C) 9

(D) 12

Answers & Solutions

Q.1 Answer:

$2^{2/3}$

Solution:

@IITJEE_Advanced

Let d be the common difference of the A.P., then

$$4 = abc = (b-d)b(b+d) = b\left(b^2 - d^2\right)$$

$$\Rightarrow b^3 = 4 + bd^2 \geq 4 \quad \left[\because b > 0, d^2 \geq 0\right]$$

$$\Rightarrow b \geq 2^{2/3}$$

Thus, minimum possible value of b is $2^{2/3}$, that is the case when $d = 0$.

Q.2 Answer:

21

Solution:

Since n^{th} term of A.P=a $+(n-1)d$

$\therefore p = 1 + (n-1)2$

$(\because$ First term $= \mathrm{a} = 1$ and common difference $= \mathrm{d} = 2)$

$\Rightarrow \quad \mathrm{n} = \frac{\mathrm{p}+1}{2}$

$\therefore (1+3+5+\ldots..+\mathrm{p}) + (1+3+5+\ldots..+\mathrm{q})$

$= (1+3+5+\ldots..+\mathrm{r})$

$\Rightarrow \frac{\frac{\mathrm{p}+1}{2}}{2}\left[2\times 1 + \left(\frac{\mathrm{p}+1}{2} - 1\right)2\right]$

$+\frac{\left(\frac{\mathrm{q}+1}{2}\right)}{2}\left[2\times 1 + \left(\frac{\mathrm{q}+1}{2} - 1\right)2\right]$

$= \frac{\mathrm{r}+1}{4}\left[2\times 1 + \left(\frac{\mathrm{r}+1}{2} - 1\right)2\right]$

$\Rightarrow \frac{\mathrm{p}+1}{4}[2 + (\mathrm{p}-1)] + \frac{\mathrm{q}+1}{4}[2 + (\mathrm{q}-1)]$

$= \frac{\mathrm{r}+1}{4}[2 + \mathrm{r} - 1]$

$\Rightarrow (\mathrm{p}+1)^2 + (\mathrm{q}+1)^2 = (\mathrm{r}+1)^2$

This is the possible only when $\mathrm{p} = 7, \mathrm{q} = 5, \mathrm{r} = 9 \therefore \mathrm{p}+\mathrm{q}+\mathrm{r} = 7+5+9 = 21$

Q.3 Answer:

$\frac{n}{n+1}$

Solution:

Let $S = \frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \ldots + \frac{1}{n(n+1)}$

Now, n^{th} term of above series $= a_n = \frac{1}{n(n+1)}$

$\Rightarrow a_n = \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ (by fraction)

Now, $\mathrm{S} = \Sigma a_n = \Sigma\frac{1}{n} - \Sigma\frac{1}{n+1}$

$= \left(1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n}\right) - \left(\frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n} + \frac{1}{n+1}\right)$

$= 1 + \left(\frac{1}{2} - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{3}\right) + \ldots + \left(\frac{1}{n} - \frac{1}{n}\right) - \frac{1}{n+1}$

$= 1 - \frac{1}{n+1} = \frac{n+1-1}{n+1} = \frac{n}{n+1}$

Q.4 Answer:

1

Solution:

@IITJEE_Advanced

$2 = 2^n \sin\alpha + \frac{1}{2^n \sin\alpha}$
$2 \cdot 2^n \sin\alpha = (2^n \sin\alpha)^2 + 1$
$(2^n \sin\alpha - 1)^2 = 0$
$\sin\alpha = \frac{1}{2^n}$
for $n = 1, \sin\alpha_1 = \frac{1}{2}$
for $n = 2, \sin\alpha_2 = \frac{1}{4}$
for $n = 3, \sin\alpha_3 = \frac{1}{8}$
$S = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \ldots$ **upto** ∞
$= \frac{\frac{1}{2}}{1-\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$

Q.5 Answer:

$\frac{xy}{x+y-1}$

Solution:

If $a, ar, ar^2, ar^3 \ldots\ldots\ldots$ **are in G.P., then**
sum of infinite G.P. $= a + ar + \ldots\ldots + \infty = \frac{a}{1-r}$
where ' a **' is the first term and '** r' **is the common ratio of GP.**
Given $x = 1 + a + a^2 + \ldots\ldots\infty$
This is a GP, with common ratio ' a **'.**
$\Rightarrow x = \frac{1}{1-a} \Rightarrow x - ax = 1 \Rightarrow a = \frac{x-1}{x}$
Again, $y = 1 + b + b^2 + \ldots\ldots\infty$ **This is also a G.P., with**
common ratio 'b'.
$\Rightarrow y = \frac{1}{1-b} \Rightarrow b = \frac{y-1}{y}$
Now, consider $1 + ab + a^2b^2 + \ldots\ldots\infty$
which is again a **GP with common ratio** ab'**.**
$\therefore$ **Sum** $= \frac{1}{1-ab} = \frac{1}{1-\frac{x-1}{x}\cdot\frac{y-1}{y}}$
$= \frac{xy}{xy-xy+x+y-1} = \frac{xy}{x+y-1}$

Q.6 Answer:

4

Solution:

Let, $S = 1 + \frac{1.3}{6} + \frac{1.3.5}{6.8} + \ldots\infty$
$\Rightarrow \frac{S}{4} = \frac{1}{4} + \frac{1.3}{4.6} + \frac{1.3.5}{4.6.8} + \ldots\infty$
$\Rightarrow \frac{1}{2} - \frac{S}{8} = \frac{1}{2} - \frac{1}{2}\cdot\frac{1}{4} - \frac{1}{2}\cdot\frac{1.3}{4.6} - \frac{1}{2}\frac{1.3.5}{4.6.8}\ldots\infty$
$\Rightarrow \frac{1}{2} - \frac{S}{8} = 1 - \frac{1}{2} + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{1.2} - \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)}{1.2.3}$
$+\frac{\frac{1}{2}\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\left(\frac{1}{2}-3\right)}{1.2.3.4}\ldots\ldots\infty$
$\Rightarrow \frac{1}{2} - S/8 = (1-1)^{1/2} = 0$
$S/8 = 1/2 \Rightarrow S = 4$

@IITJEE_Advanced

Q.7 Answer:

$6\left(\frac{3^n}{2^n}-1\right)-2n$

Solution:

$t_n = S_n - S_{n-1} = 3t_n - 3t_{n-1} \Rightarrow \frac{t_n}{t_{n-1}} = \frac{3}{2}$

Hence the series is geometric with common ratio $\frac{3}{2}$**. Also,** $S_1 = 3t_1 - 2 \Rightarrow t_1 = 3t_1 - 2 \Rightarrow t_1 = 1$

Now, $S_n = \frac{a(r^n-1)}{r-1} = \frac{\left(\frac{3}{2}\right)^n - 1}{\frac{3}{2}-1} = 2\left\{\left(\frac{3}{2}\right)^n - 1\right\}$

$\therefore S_1 + S_2 + S_3 + \ldots\ldots + S_n$

$= 2\left[\begin{array}{c}\left\{\frac{3}{2}-1\right\} + \left\{\left(\frac{3}{2}\right)^2 - 1\right\} + \\ \left\{\left(\frac{3}{2}\right)^3 - 1\right\} + \ldots. + \left\{\left(\frac{3}{2}\right)^n - 1\right\}\end{array}\right]$

$= 2\left[\left\{\frac{3}{2} + \left(\frac{3}{2}\right)^2 + \left(\frac{3}{2}\right)^3 + \ldots\ldots\ldots\ldots + \left(\frac{3}{2}\right)^n\right\} - n\right]$

$= 2\left[\frac{\frac{3}{2}\left\{\left(\frac{3}{2}\right)^n - 1\right\}}{\frac{3}{2}-1} - n\right] = 2\left[3\left\{\left(\frac{3}{2}\right)^n - 1\right\} - n\right]$

$= 6\left(\frac{3^n}{2^n} - 1\right) - 2n$

Q.8 Answer:

990

Solution:

Let $S_1 = 2 + 3 + 5 + 9 + 16 + \ldots\ldots\ldots\ldots + x_n$

$\frac{S_1 = 2+3+5+9+\ldots\ldots\ldots x_{n-1}+x_n}{O = 2+[1+2+4+7+\ldots\ldots}$

+ to $(n-1)$ **term** $] - x_n$

$\therefore x_n = 2 + [1 + 2 + 4 + 7 + \ldots$ **to** $(n-1)$ **terms** $]$

Again let $S_2 = 1 + 2 + 4 + 7 + \ldots\ldots\ldots + t_{n-1}$

$\frac{S_2 = 1+2+4+7\ldots+t_{n-2}+t_{n-1}}{O = 1+[1+2+3+\ldots\ldots\ldots+(n-2)\text{ term }]-t_{n-1}}$

$t_{n-1} = 1 + \frac{(n-2)(n-1)}{2} = \frac{n^2-3n+4}{2}$

$\therefore S_2 = \sum_{n=1}^{n-1} t_{n-1} = \frac{1}{2}\Sigma n^2 - \frac{3}{2}\Sigma n + 2\Sigma 1$

$= \frac{1}{2}\frac{(n-1)n(2n-1)}{6} - \frac{3}{2}\frac{n(n-1)}{2} + 2(n-1)$

$= (n-1)\left[\frac{2n^2-n}{12} - \frac{3n}{4} + 2\right]$

$= \frac{n-1}{12}\left[2n^2 - n - 9n + 24\right]$

$= \frac{n-1}{6}\left[n^2 - 5n + 12\right] = \frac{n^3-6n^2+17n-12}{6}$

$\therefore x_n = 2 + S_2 = 2 + \frac{n^3-6n^2+17n-12}{6}$

$= \frac{n^3-6n^2+17n}{6}$

So, $x_{20} = 990$

Q.9 Answer:

2

Solution:

@IITJEE_Advanced

Let S_n denote the sum of n terms of an A.P. According to given

$\frac{S_n}{S_{2n}-S_n} = k \quad \forall n \geq 1 \Rightarrow \frac{S_1}{S_2-S_1} = \frac{S_2}{S_4-S_2}$

$\Rightarrow S_1\, S_4 - S_1\, S_2 = S_2^2 - S_1\, S_2 \Rightarrow S_1\, S_4 = S_2^2$

$\Rightarrow a\frac{4}{2}[2a + (4-1)d] = (a + a + d)^2$

$\Rightarrow a(4a + 6d) = (2a + d)^2$

$\Rightarrow 2ad = d^2 \Rightarrow 2a = d$

Since $a = 1$, we get $d = 2$

Q.10 Answer:

$\frac{n}{n+1}$

Solution:

The general term is

$T_n = \frac{\frac{n}{2}.\frac{n+1}{2}}{1^3+2^3+3^3+\ldots+n^3} = \frac{1}{n(n+1)}$

$= \frac{1}{n} - \frac{1}{n+1}$

$\therefore \quad S_n = 1 - \frac{1}{n+1} = \frac{n}{n+1}$

Q.11 Answer:

G.P. with common ratio $e^{d/\left(b^2-d^2\right)}$

Solution:

a, b, c **are in A.P.** $\Rightarrow 2\, b = a + c$

Now,

$e^{1/c} \times e^{1/a} = e^{(a+c)/ac} = e^{2\, b/ac} = \left(e^{b/ac}\right)^2$

$\therefore e^{1/c}, e^{b/ac}, e^{1/a}$ **in G.P. with common ratio**

$= \frac{e^{b/ac}}{e^{1/c}} = e^{(b-a)/ac} = e^{d/(b-d)(b+d)}$

$= e^{d/\left(b^2-d^2\right)}$ [$\because$ **a, b, c are in A.P. with common difference** $d \therefore b - a = c - b = d$

Q.12 Answer:

21

Solution:

$S = \frac{1}{1-\frac{1}{2}} = 2$

$S_n = \frac{1\left(1-\frac{1}{2^n}\right)}{1-\frac{1}{2}} = 2 - \frac{1}{2^{n-1}}$

Given $S - S_n < 10^{-6}$

$\therefore \quad \frac{1}{2^{n-1}} < 10^{-6} \Rightarrow 2^{n-1} > 10^6$

$\therefore \quad n - 1 > 6 \log_2 10 = \frac{6}{0.3010} n > 20$

$\left[\because \frac{6}{.3018} < 20\right.$

$n = 21$

@IITJEE_Advanced

Q.13 Answer:

n^2

Solution:

$S_n = 1 + 2a + 3a^2 + ... + na^{n-1}$

or $aS_n = a + 2a^2 + ... + (n-1)a^{n-1} + na^n$

where, $a = 1 + \frac{1}{n}$

$\therefore (1-a)S_n = 1 + a + a^2 + ... + a^{n-1} - na^n$

$(1-a)S_n = \frac{a^n-1}{a-1} - na^n$

$\Rightarrow -\frac{1}{n}S_n = \frac{\left(1+\frac{1}{n}\right)^n-1}{\frac{1}{n}} - n\left(1+\frac{1}{n}\right)^n = -n$

$\Rightarrow S_n = n^2$

Q.14 Answer:

None of the above series

Solution:

Given series is $1 + \frac{1}{\sqrt{3}} + 3 + \frac{1}{3\sqrt{3}} + ...$

Consider $\frac{a_2}{a_1} = \frac{1}{\sqrt{3}}, \frac{a_3}{a_2} = \frac{3}{1/\sqrt{3}}, \frac{a_4}{a_3} = \frac{1/3\sqrt{3}}{3}$

Also find $a_2 - a_1 = \frac{1}{\sqrt{3}} - 1 = \frac{1-\sqrt{3}}{\sqrt{3}}$

$a_3 - a_2 = 3 - \frac{1}{\sqrt{3}} = \frac{3\sqrt{3}-1}{\sqrt{3}}$

$a_4 - a_3 = \frac{1}{3\sqrt{3}} - 3 = \frac{1-9\sqrt{3}}{3\sqrt{3}}$

Thus, $\frac{a_2}{a_1} \neq \frac{a_3}{a_2} \neq \frac{a_4}{a_3}$

and $a_2 - a_1 \neq a_3 - a_2 \neq a_4 - a_3$

Hence, given series is neither A.P., G.P. nor H.P.

Q.15 Answer:

GP.

Solution:

Let the 3 n terms of G.P. be a, ar, ar

$^2, ... ar^{n-1}, ar^n, ar^{n+1}, ... ar^{2n-1}, ar^{2n}, ar^{2n+1}, ...$

ar^{3n-1}**. Then**

$S_1 = a + ar + ar^2 + ... + ar^{n-1} = \frac{a(1-r^n)}{1-r}$

$S_2 = ar^n + ar^{n+1} + ... + ar^{2n-1} = \frac{ar^n(1-r^n)}{1-r}$

$S_3 = ar^{2n} + ar^{2n+1} + ... + ar^{3n-1} = \frac{ar^{2n}(1-r^n)}{1-r}$

Clearly $\frac{S_2}{S_1} = \frac{S_3}{S_2} = r^n$

Q.16 Answer:

$\leq \frac{1}{2n+1}$

Solution:

@IITJEE_Advanced

$x+\frac{1}{x} \geq 2, \ldots, x^n+\frac{1}{x^n} \geq 2$
in adding $\left(x+\frac{1}{x}\right)+\left(x^2+\frac{1}{x^2}\right)+\ldots+\left(x^n+\frac{1}{x^n}\right) \geq 2n$
$\left(\frac{1}{x^n}+\frac{1}{x^{n-1}}+\ldots \frac{1}{x}\right)+1+\left(x+x^2+\ldots+x^n\right) \geq 1+2n$
$\frac{\left(1+x+\ldots+x^{n-1}+x^n\right)+x^{n+1}+x^{n+2}+\ldots+x^{2n}}{x^n}$
$\geq 1+2n$
$\frac{x^n}{1+x+\ldots+2^{2n}} \leq \frac{1}{1+2n}$

Q.17 Answer:

$\sqrt{m}+\sqrt{m-1}: \sqrt{m}-\sqrt{m-1}$

Solution:

We have $\frac{a+b}{2}=m \frac{2ab}{a+b}$
or $\frac{(a+b)^2}{4ab}=m$
or $\frac{(a+b)^2}{4ab}-1=m-1$
or $\frac{(a-b)^2}{4ab}=m-1$
Dividing (i) by (ii), we get $\frac{(a+b)^2}{(a-b)^2}=\frac{m}{m-1}$ or $\frac{a+b}{a-b}=\frac{\sqrt{m}}{\sqrt{m-1}}$
Applying componendo and dividendo, We get $\frac{\mathrm{a}}{\mathrm{b}}=\frac{\sqrt{\mathrm{m}}+\sqrt{\mathrm{m}-1}}{\sqrt{\mathrm{m}}-\sqrt{\mathrm{m}-1}}$

Q.18 Answer:

9

Solution:

$\mathrm{AM} \geq \mathrm{HM}$
$\Rightarrow \frac{\mathrm{a}+\mathrm{b}+\mathrm{c}}{3} \geq \frac{3}{\frac{1}{\mathrm{a}}+\frac{1}{\mathrm{b}}+\frac{1}{\mathrm{c}}}$
$\Rightarrow (a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$

Q.19 Answer:

$\frac{3^{10} \cdot 2^4}{7^7}$

Solution:

Taking A.M. and G.M. of 7 numbers $\frac{\mathrm{a}}{2}, \frac{\mathrm{a}}{2}, \frac{\mathrm{b}}{3}, \frac{\mathrm{b}}{3}, \frac{\mathrm{b}}{3}, \frac{\mathrm{c}}{2}, \frac{\mathrm{c}}{2}$, we get
$\frac{2 \cdot \frac{\mathrm{a}}{2}+3 \cdot \frac{\mathrm{b}}{3}+2 \cdot \frac{\mathrm{c}}{2}}{7} \geq\left\{\left(\frac{\mathrm{a}}{2}\right)^2\left(\frac{\mathrm{b}}{3}\right)^3\left(\frac{\mathrm{c}}{2}\right)^2\right\}^{\frac{1}{7}}$
$\Rightarrow \quad \frac{3^7}{7^7} \geq \frac{a^2 b^3 c^2}{2^2 \cdot 3^3 \cdot 2^2} \quad \Rightarrow \quad a^2 b^3 c^2 \leq \frac{3^{10} \cdot 2^4}{7^7}$
$\therefore \quad$ greatest value of $a^2 b^3 c^2=\frac{3^{10} \cdot 2^4}{7^7}$

Q.20 Answer:

3

Solution:

@IITJEE_Advanced

Given expression is $\frac{1}{2}\sum\frac{2a}{b+c-a}$

$=\frac{1}{2}\sum\left(\frac{2a}{b+c-a}+1\right)-\frac{3}{2}=\frac{1}{2}(a+b+c)$

$\sum\frac{1}{b+c-a}-\frac{3}{2}$

Now, as $(a+b+c)=\sum(b+c-a)$

Applying A.M. $\geq$ **H.M.**

Minimum value of the expression $=\frac{1}{2}\times 9-\frac{3}{2}=3.$

@IITJEE_Advanced